Lab_01_ArtimeticExpression

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0.1 Content Learning Objectives

After completing this activity, students should be able to:

- Execute mathematical expressions similar to a calulator
- Describe the functions of the three Python division operators
- Explain the differences between integers and floating-point data

0.2 Processing Skill Goals

During the activity, students should make progress towards:

- Recognizing mathematical operations based on tables (Information Processing)

0.3 Atmospheric Sciences Concepts

Scientific programming is a very important skill for this field. This class will set the basis for all of your future programming. These skills will be essential for classes like *Computer Concepts in Meteorology* (ATSC 270), *Remote Sensing of the Atmosphere* (ATSC 345), *Physical Meteorology* (ATSC 353), etc. Pretty much every ATSC class after ATSC 110 will have some form of scientific programming involved.

This means that it is important you understand the basic of computer science and coding with Python. This will make upper level classes much easier, less stressful, and less time consuming. Actually learning the basics now will set you up to be as successful as possible later.

0.4 Importance to Atmospheric Sciences

Math is the foundation of all Atmospheric Sciences and is used for:

- Temperature Conversions
- The Ideal Gas Law
- Wind Speed Conversions
- Wind Direction Conversions
- Atmospheric Dynamics
- Potential Temperature Calculations
- Numerical Weather Prediction
- Much, much more!!!

1 Part 1 Python Calculator

Open a Jupyter Console by clicking the "+" next to the Lab_01.ipynb tab at the top of the screen. Next click the "Python 3 (ipykernal)" under "Console" to start the Console.

There will be an empty textbox on the bottom of the screen. This textbox has a flashing cursor as a prompt indicating that the interpreter is waiting for input. All text entered after the ***prompt*** will be executed immediately as Python code.

If you type a Python ***expression*** (code that results in a value) after the prompt, Python will show the value of that expression, similar to a calculator. You can use Python's *math* module to perform more complex mathematical operations like logarithms and trigonometric operations.

1.0.1 Read the questions first! Do not add to the table yet!

Python Code	Predicted Output	Actual Output
2 + 3	5	5
3 * 4 + 2	14	14
3 * 4 + 2.0	14	14.0
3(4 + 2)	18	TypeError
3 * (4 + 2)	18	18
5 / 10	0.5	0.5
5 / 10.0	0.5	0.5
5 / 9	5/9	0.555555556
2 ** 4	8*	16
abs(-2) ** 4	8*	16
math.pow(2,4)	error	NameError
import math	nothing*	nothing
math.pow(2,4)	16	16.0
sqrt(4)	error	NameError
math.sqrt(4)	2	2.0
math.cos(0)	1*	1.0
math.pi	3.1415926 etc.	3.1415926535
math.sin(math.pi / 2)	1*	1.0

1.1 Questions test

1. In the middle "Predicted output" column, write what value you expect will be displayed, based on your experience using a calculator. If there are any lines you are not confident about, place an asterisk next to your predicted output.

2. In the Jupyter Console type each Python expression at the prompt, one line at a time, and write the corresponding Python output in the third column above. If an error occurs, write what type of error it was (i.e., the first word of the last line of the error message).

3. What does the ** operator do?

   It acts as an exponent.

4. Based on the Python code in Part 1, identify four examples of:

   a) mathematical operator + ,** ,*, /

   b) mathematical function math.pow, math.sqrt, math.cos, math.sin

5. For addition and multiplication to produce an output with a decimal value, what type of number must be part of the input? Provide justification for your answer.

   You must include a decimal value in your imput because otherwise the progra

6. Does division follow the same rule as in #5? Provide justification for your answer.

   No, because even if you divide a number that doesn't become a fraction, like 1

7. The output of Part 1 displayed three different errors. Explain the reason for each:

   a) TypeError

   There was no mathematical operator to tell python to multiply three by four plu

   b) 1st NameError

   math had not been imported, so python didn't know what that meant.

   c) 2nd NameError

   math was not put in front of sqrt, so python didn't recognize the function. We d

8. Identify two differences between using a Python built-in function (e.g., abs) and a function from the math module.

   To use a function from the math module, you must import math and type it in t

1.2 Atmospheric Sciences Questions

1a. To convert temperatures from Farenheit to Celsius the equation c = 5/9 * (F - 32) is used. Use the console to convert from Farenheit to Celsius. Fill the table with your results rounded to 2 decimals. Put all of the equations you used in the text box below.

1b. Converting from Celsius to Farenheit uses the equation F = (c * 9/5) + 32. Use the console to convert from Celsius to Farenheit. Fill the table with your resutls rounded to 2 decimals. Put all of the equations you used in the text box below.

```
print(5/9*(32-32))        print(5/9*(100-32))        print(5/9*(212-32))

print(5/9*(-40-32))       print(5/9*(57.6-32))       print(5/9*(14.9-32))

print((32*9/5)+32)        print((20.5*9/5)+32)       print((-10*9/5)+32)

print((13*9/5)+32)        print((0*9/5)+32)          print((57.6*9/5)+32)
```

Farenheit	Celsius
32	0.00
100	37.78
212	100.0(
-40	-40.00
57.6	14.22
14.9	-9.50
89.60	**32**
68.90	**20.5**
14.00	**-10**
55.40	**13**
32.00	**0**
135.68	**57.6**

The Ideal Gas Law is the equation of state of a hypotetical ideal gas. It is regularly used to describe the behavior of the atmosphere. It is made up of 5 variables: P = Pressure (Pa), V = Volume (m^3), n = Number of Moles (mol), R = Gas Constant (8.314 m^3 Pa K^{-1} mol^{-1}), T = Temperature (^{o}K). The Ideal Gas Law equation can be rearranged to find any of these variables.

PV =nRT

2. Using the console, use the Ideal Gas Law equation to fill in the table rounding answers to 2 decimal points. Put all of the equations you used in the textbox below.

```
print((1013*100)/(8.314*30)), print((2164*8.314*17)/850), print((4826*8.314*-10)/12(
```

Pressure	Volume	Number of Moles	Temperature
1013	**100**	406.14	**30**
850	359.8	**2164**	**17**
-3343	**120**	**4826**	**-10**
300	**90**	**1483**	2.19
1000	**100**	**4000**	3.01
850	**100**	**4000**	2.56
500	**100**	**4000**	1.50
300	**100**	**4000**	0.90

2 Part 2 Dividing Numbers

Table A	Table B	Table C
9 / 4 = 2.25	9 // 4 = 2	9 % 4 = 1
10 / 4 = 2.5	10 // 4 = 2	10 % 4 = 2
11 / 4 = 2.75	11 // 4 = 2	11 % 4 = 3
12 / 4 = 3.0	12 // 4 = 3	12 % 4 = 0
13 / 4 = 3.25	13 // 4 = 3	13 % 4 = 1
14 / 4 = 3.5	14 // 4 = 3	14 % 4 = 2
15 / 4 = 3.75	15 // 4 = 3	15 % 4 = 3
16 / 4 = 4.0	16 // 4 = 4	16 % 4 = 0

2.1 Questions

9. For each operator in Part 2, identify the symbol and describe the type of numerical result.

   / is a division operator that gives a float. // Is a division operator that gives the resu

10. If the result of the / operator were rounded to the nearest integer, would this be the same as the result of the // operator? Explain how the results in Table A compare to Table B.

    No, because when we round, we look to the integer to the right and if it is five or gı

11. If the table included more rows, list all numbers // 4 would evaluate to 2 and all the numbers // 4 would evaluate to 4.

    17/4 evaluates to 4.25, 17//4 evaluates to 4, and 17%4 evaluates to 1. 18/4 evalua

12. Based on the results of Table C, propose another number % 4 evaluates to 0, and explain what all these numbers have in common.

    20%4 evaluates to 0 because 4 is a multiple of 20 and 20 divided by 4 gives us an

13. Consider the expressions in Table C that evaluate to 1. How do the left operands in these expressions (i.e., 9, 13) differ from those that evaluate to 0?

Four is not a multiple of these numbers, so you cannot divide them evenly by 4 an

14. Describe the reason for the repeated sequence of numbers (0, 1, 2, 3) for the result of % 4.

Because there are four remainder options that you can get when you divide by 4: (

15. Recall how you learned to do long division in elementary school. Finish solving for 79 / 5 below. What is 79 / 5? Which part of the answer is 79 // 5, and which part is 79 % 5?

79/5 evaluates to 15.8. The 15 is the part of the answer that equals 79//5, and the

16. Imagine that you are given candy mints to divide evenly among you and two friends.

a) If you receive 11 mints, how many mints would you and your friends each get, and how many are left over? Write a Python expression to compute each result.

We would each receive 3 mints, and there would be 2 mints left over. The python e

b) If you receive 2 more mints, how many mints would you and your friends each get, and how many are left over? Write a Python expression to computes this result.

'/With two additional mints, we would all get 4 mints, and there would be 1 left over

17. Python has three division operators: "floor division", "remainder division", and "true division". Which operator (symbol) corresponds to each name?

True division corresponds to /, remainder division corresponds to %, so I assume t

2.2 Atmospheric Sciences Questions

Weather stations around the world collect information like temperature, pressure, humidity, wind speed, and widn direction amoung other things. Wind direction is typically recorded in degrees from 0 to 360. Occasionally, wind direction will be reported as being greater than 360 degrees. If you are ever working with a data set where this happens it is important to correct for this error. (North winds can be reported as 0 or 360)

3. Using one of the division operators, write a code to find the actual wind direction from 0 to 360 degrees and fill in the table with the correct wind directions. (Hint: you will need to do some multiplication and substraction after using one of the division operators)

wind = (whatever integer) (new line) print(wind%360)

Large Wind Direction	Actual Wind Direction
720	0
800	80
1516	76
1000	280
7843	283
1254	154

3 Part 3 Integers and Floats

Each value in Python has a *data type* which determines what can be done with that data. Enter the following code, one line at a time, into the Jupyter Console. Record the output for each line (if any) in the second column.

Python Code	Console Output
integer = 3	no output
type(integer)	<class 'int'>
type("integer")	<class 'str'>
pi = 3.1415	no output
type(pi)	<class 'float'>
word = str(pi)	no output
word	'3.1415'
number = float(word)	no output
print(word * 2)	3.14153.1415
print(number * 2)	6.283
print(word + 2)	TypeError
print(number + 2)	5.1415
euler = 2.7182	no output
int(euler)	2
round(euler)	3

3.1 Questions

18. What is the data type (int, float, or str) of the following values?

    a) pi float

    b) integer int

    c) word str

    d) number float

19. List the function calls that convert a value to a new data type.

    str(), float(), int()

20. How does the behavior of the operators (+ and *) depend on the data type?

    You cannot manipulate the numbers mathematically if they are strings; they must be i

21. What is the difference between the int function and the round function?

    The int function just takes the whole number that is a part of your float, while the roun

22. What is the value of 3 + 3 + 3? What is the value of .3 + .3 + .3? If you enter these expressions into a Python Shell, what do you notice about the results?

    3+3+3 equals 9, while .3+.3+.3 equals .9. But if I enter this in python, 3+3+3 remains t

23. In order to store a number with 100% accuracy, what data type is required? How might you precisely represent a bank account balance of $123.45?

    A string, so that the computer doesn't store the number incorrectly. Then it can be co

24. Try calculating a very large integer in a Python Shell, for example, 123^{456}. Is there a limit to the integers that Python can handle?

    Python gave me an integer answer with 953 characters in it, so I assume that there m

25. Try calculating a very large floating-point number in a Python Shell, for example, 123.0^{456}. Is there a limit to the floating-point numbers that Python can handle?

    The result was finally too large and gave me an OverflowError, so yes we have now d

26. Summarize the difference between the numeric data types (int and float). What are their pros and cons?

    Python seems to be able to handle larger ints, but ints can be less accurate than float

3.2 Atmospheric Sciences Questions

4. Identify which variables should be floats, ints, or strings in the table below. If you are unsure put * after your answer.

Variable	Data Type
Temperature	float
Wind Speed	int*
Humidity	float
Wind Direction (Degrees)	int

Variable	Data Type
Wind Direction (Cardinal Direction)	string
Pressure	int*
Distance	float
Cloud Cover	float or string
Current Weather	string
Date	string